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Smart Reservoir Management and Proactive Model Enhancement with Ultra-Deep Azimuthal Resistivity: From Real-Time Mapping to Post-Drilling Applications

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Abstract

This paper presents the transformative application of Ultra-deep Azimuthal resistivity (UDAR) data beyond geosteering, highlighting its value in post drilling workflows such as smart reservoir management and enhanced reservoir modeling. Post drilling UDAR reservoir data is leveraged to map the oil-water contact and reservoir boundaries, enabling the construction of a three-dimensional view of the remaining oil column. This enables the refinement of dynamic reservoir models and strategically plan infill wells to maximize reservoir contact and continue maintaining optimum stand-off from water.

The latest multi-dimensional 3D UDAR technology acquires a full 360 electromagnetic tensor measurements in real time, generating high resolution resistivity volumes with depth of investigation exceeding 100 feet. The workflow includes 3D structural interpretation, geobodies extraction and delivering 3D resistivity volumes in SEG-Y format, following Society of Exploration Geophysicists (SEG) standard.

We demonstrate how UDAR data supports five key post drilling domains: reservoir characterization, intelligent well placement, completion design optimization, history match validation and 4D surveillance modeling. Field examples illustrate how UDAR driven insights improve confidence in oil-water contact mapping and enhancing surveillance density to improve fluid distribution certainty.

Our findings position UDAR as a critical enabler of smart reservoir management. Its integration into post drilling workflows improves subsurface understanding and supports more informed development planning. As UDAR evolves alongside machine learning and digital twin technologies, it paves the way for real time, predictive modeling: unlocking smarter, faster and more sustainable decision making in complex reservoirs.

Introduction

UDAR has changed the game for horizontal well geosteering, especially in reservoirs with geological heterogeneity, stratigraphic complexity and/or challenging reservoir dynamics. UDAR tools, with their ability to measure directional resistivity tens to hundreds of feet from the wellbore, give critical information to maximize reservoir contact and keep maximum standoff from fluid contacts such as oil-water. These measurements play a critical role in avoiding early water breakthroughs and maximize hydrocarbon production.

While the real-time benefits of UDAR in geosteering are well documented in the literature, its value goes far beyond the drilling phase. The high-resolution spatial resistivity data captured during drilling can be reinterpreted and integrated into post-drilling workflows to create more value and enhance reservoir understanding and inform long term field development. This paper will explore five key post-drilling domains where UDAR data can deliver technical and economic value: improved reservoir characterization, proactive completion design, multi-dimensional mapping for intelligent well placement, subsurface history match validation and 4D time-lapse surveillance model enhancement. By deploying UDAR data across these domains, operators get an integrated multi-scale understanding of the reservoir, improve both geological and dynamic certainty and make higher confidence development decisions. These additional applications allow for a more holistic field development by linking real-time mapping data to longer term reservoir behavior.

Well placement is challenging in a geologically complex reservoir. For a well to be considered successful, operators must use all available historical data to determine the production life of the well. It becomes more challenging when more data is needed to represent the current fluid dynamic state of the field. In such cases, well planning relies heavily on real-time reservoir characterization, including properties distribution and structural framework around the wellbore to bridge gaps in reservoir understanding and optimize well placement to maximize hydrocarbon production over the longest duration possible.

The latest advancements in reservoir mapping-while-drilling technology have enabled a more complete understanding of the reservoir with its multi-dimensional reservoir scale mapping capabilities, minimizing the need for pilot holes. Fluids distribution around the wellbore is a major factor in determining the life of an oil producer. Using UDAR technology along with logging-while-drilling systems is critical to guide the completions design to ensure uniform sweep across the horizontal section. This shows how multi-dimensional UDAR system converts ultra-deep electromagnetic measurements into resistivity profile around the borehole, high-definition borehole resistivity images and formation pressure measurements helped to confirm individual flow units.

Ultra-Deep Azimuthal Resistivity Mapping Technology

UDAR technology has become a cornerstone of modern reservoir mapping-while-drilling operations, providing critical insights for geosteering, well placement, and real-time reservoir characterization. UDAR tools represent a significant advancement in directional resistivity logging, designed to provide detailed 3D images of the subsurface while drilling. This capability allows operators to map reservoir layers and fluid boundaries beyond the immediate drilling area with high resolution.

Evolution of Ultra-Deep Azimuthal Resistivity Technology

Resistivity logging technology has significantly evolved over time. Early tools measured resistivity very close to the borehole and assumed the surrounding rock was uniform. This limited their ability to detect complex reservoir features. Over time, azimuthal resistivity tools were introduced. These tools measure resistivity in different directions around the borehole, helping to identify features like dipping beds and fractures. But even these had limitations in depth and resolution. The breakthrough came with full electromagnetic (EM) tensor measurements. Modern UDAR tools use multiple transmitters and receivers arranged to capture the complete EM field, including magnitude, phase, and direction. This rich dataset allows detailed imaging of the reservoir's structure and fluid contacts in three dimensions. At the same time, inversion techniques evolved from simple 1D models to advanced multi-dimensional inversion algorithms. These algorithms solve complex mathematical problems to create detailed 3D resistivity models that match the measured data. With advancements in cloud computing, these inversions can now be performed quickly in real time while drilling. [Figure 1](#) shows the evolution of UDAR over time

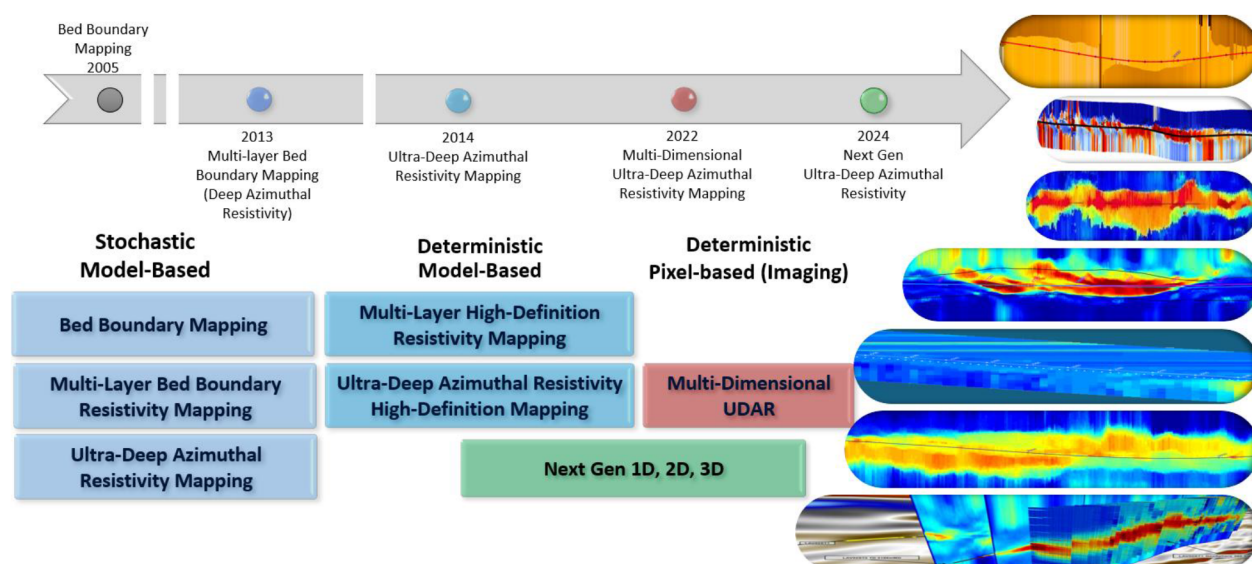


Figure 1—Evolution of Ultra-deep Azimuthal resistivity (UDAR) technology with time

How the UDAR Tool Works

Through a modular system of transmitters and multifrequency receivers embedded in the bottomhole assembly, UDAR tools enable directional deep resistivity measurements that can probe tens to hundreds of feet into the formation, far beyond the reach of conventional logging-while-drilling techniques (Beer et al., 2010; Netto et al., 2012). These tools reduce reliance on pilot holes and improve operational efficiency by enabling real-time decisions based on accurate imaging of formation structures and fluid boundaries. The core advantage of UDAR lies in its capability to resolve three-dimensional resistivity contrasts around the borehole while drilling. The system typically consists of a transmitter and up to three independent receiver modules spaced along the bottomhole assembly as shown in Figure 2. Each receiver operates at multiple frequencies commonly 2 kHz, 6 kHz, 12 kHz, 24 kHz, 48 kHz, and 96 kHz—allowing for flexibility in configuration and optimization of signal-to-noise ratios depending on the surrounding formation resistivity and desired depth of investigation. This multifrequency capability makes it possible to acquire robust electromagnetic data across a wide range of geological settings (Seydoux et al., 2014).

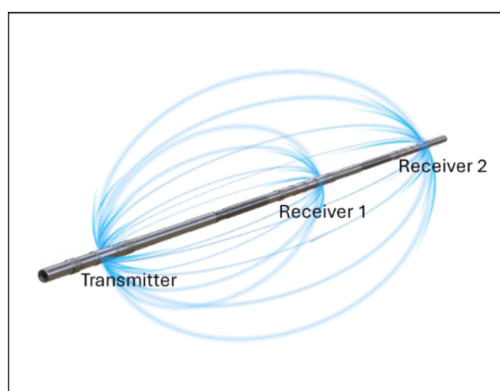


Figure 2—A borehole assembly with a UDAR transmitter and 2 UDAR receivers showing volumetric signal sensitivity to the formation surrounding the wellbore

The data acquisition process involves measuring the full resistivity tensor, which provides a detailed, azimuthally sensitive image of the subsurface. The tensor measurements capture anisotropic effects and directional resistivity variations that are often critical in complex reservoirs. With transmitter-receiver

spacings up to 100 feet, the effective depth detection of UDAR can reach similar radial distances from the borehole, although the effective depth is influenced by formation resistivity and tool configuration. The system can be configured to operate at three optimal frequencies out of the six available, with selections made during pre-job planning based on anticipated formation characteristics. A typical UDAR tool assembly for all subject wells discussed in this paper is shown in Figure 3

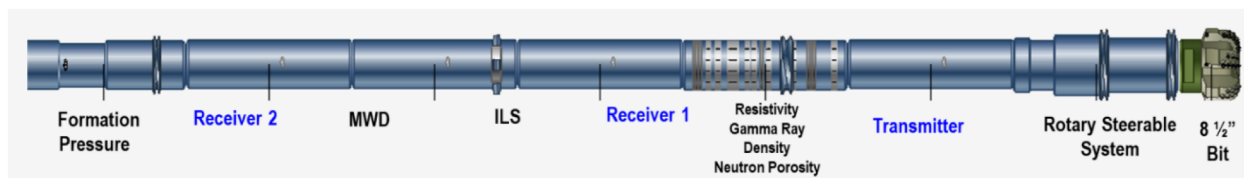


Figure 3—A borehole assembly with a UDAR transmitter and 2 UDAR receivers deployed for well discussed in this paper.

Processing and Inversion

Once the raw data is acquired, it undergoes several processing steps, including noise filtering and corrections for borehole effects and tool positioning. Inversion algorithms are central to the interpretation of UDAR measurements. Earlier implementations relied on 1D stochastic and deterministic inversions that assumed formation bedding was laterally homogeneous and parallel to the wellbore trajectory (Dong et al., 2015; Wilding-Steele et al., 2019). These algorithms produced 2D resistivity curtain sections by generating inversion points at fixed intervals along the well path, typically every one meter. This approach provided valuable insights for low-complexity formations but proved inadequate in more geologically intricate environments where assumptions of lateral homogeneity did not hold. To address these limitations, newer UDAR systems incorporate multi-dimensional inversion frameworks. The transition from 1D to 2D inversion was a significant advancement, enabling transverse mapping of resistivity features and resolving lateral heterogeneities (Thiel and Omeragic, 2018, Thiel and Omeragic, 2019). The inversion process uses components of the full electromagnetic tensor and inverts for horizontal and vertical resistivity values, as well as the angular relationship between the wellbore and formation structure. This enhanced capability allows for improved mapping of thin beds, shale barriers, and laterally varying fluid contacts, offering a more detailed understanding of the surrounding geology and fluid level identification.

The latest developments have pushed UDAR technology into full 3D interpretation. By leveraging pixel-based inversion methods and cloud computing, real-time multi-dimensional resistivity volumes can now be derived while drilling (Thiel et al., 2021). Adaptive regularization is employed to stabilize the inversion and preserve resolution, particularly in areas with steep resistivity gradients or intricate structural configurations. To validate these multi-dimensional inversion workflows, synthetic models simulating realistic geological scenarios have been extensively used. These include cases with laterally displaced sand bodies, multifingered injectites, and shale intrusions (Wang et al., 2021). The results have demonstrated the system's ability to quantify the size, orientation, and resistivity contrast of features around the wellbore, outperforming previous generation systems in both accuracy and resolution. The integration of cloud-based inversion platforms also allows for the rapid processing of high-volume UDAR data streams. By transmitting compressed measurement data in real time, and distributing inversion workloads across scalable computing resources, real-time reservoir characterization becomes both feasible and practical. This capability is essential for real time, proactive geosteering, especially in high-angle or horizontal wells where quick decisions are required to maintain optimal trajectory within a productive zone. Automated inversion engines used in UDAR systems operate independently of user assumptions, relying solely on the data acquired downhole. Depending on the reservoir complexity and operational objectives, different inversion approaches—stochastic, deterministic, or pixel-based—can be executed in parallel to maximize interpretive robustness. Stochastic inversions offer uncertainty quantification by generating a range of models that fit the

acquired data, making them valuable in low signal-to-noise environments (Dong et al., 2015). Deterministic inversions, on the other hand, resolve more layers and offer higher spatial resolution, making them suitable for mapping fine stratigraphic features (Wilding-Steele et al., 2019).

To further refine inversion results, measurements from various receiver spacings are processed both individually and collectively. Long-spacing receivers provide deeper penetration but lower vertical resolution, while short-spacing receivers capture finer details closer to the borehole. Processing these datasets in combination enhances both the vertical and lateral resolution of the resistivity map. This multi-scale inversion approach enables more accurate interpretation of formation boundaries, fluid contacts, and geological heterogeneities. In high-definition configurations, UDAR systems are enhanced with more powerful transmitters and increased spacing—up to 40 meters—to extend detection range and improve signal quality (Xu et al., 2018). These improvements further support the delineation of thin and deeply located geological features, expanding the applicability of UDAR beyond conventional geosteering into reservoir characterization and completions planning. Overall, UDAR technology continues to redefine subsurface imaging by bridging the gap between logging-while-drilling measurements and full-field reservoir models. By offering real-time, high-resolution mapping of complex geology in three dimensions, UDAR empowers drilling and reservoir engineers to make better-informed decisions, ultimately improving hydrocarbon recovery and reducing drilling risk.

Post drilling Applications of UDAR data

In the realm of reservoir engineering and well planning, the integration of advanced technologies and state of art inversion results is crucial for optimizing well placement decision and enhancing reservoir characterization. The following examples illustrate the application of UDAR tools and multidimensional reservoir mapping techniques in achieving precise well placement and comprehensive reservoir understanding. These examples highlight the significance of detailed structural interpretations and real-time geosteering in navigating complex depositional environments and ensuring successful wellbore trajectories.

Improved Reservoir Characterization. In this example, the subject well ‘X’ was strategically planned to target a specific sand layer within a multilayer reservoir package characterized by a fluvial depositional system. The primary objective was to accurately place the well within the target zone, which required the use of an UDAR tool to further support in well placement and reservoir mapping. The results, depicted in Figure 4, reveal the multilayer mapping of various sand bodies above and below the wellbore, including a distinct 5 ft thin intra-shale layer separating the target sand from another resistive sand body below. Additionally, the tool confirmed the identification of the oil water contact at a safe distance from the wellbore.

To further enhance reservoir characterization, advanced answer products such as 3D resistivity volumes and geobodies were generated and interpreted using legacy seismic interpretation tools. This provided a 3D structural interpretation of the different layers observed in the multidimensional resistivity inversion. A detailed analysis indicated that the target sand body was pinching out to the south of the wellbore, and if the wellbore had been planned just a few feet to the south of its current trajectory, the wellbore would have drilled a significant amount of shale without understanding that the sand had disappeared. The 3D resistivity volume was also utilized to visualize the pay zone by applying a resistivity cutoff and producing geobodies that could be integrated into existing geomodeling workflows. The structural interpretations were instrumental in updating the existing grids, capturing the depth and morphology of the encountered sands and meandering sand channels typical of this depositional environment.

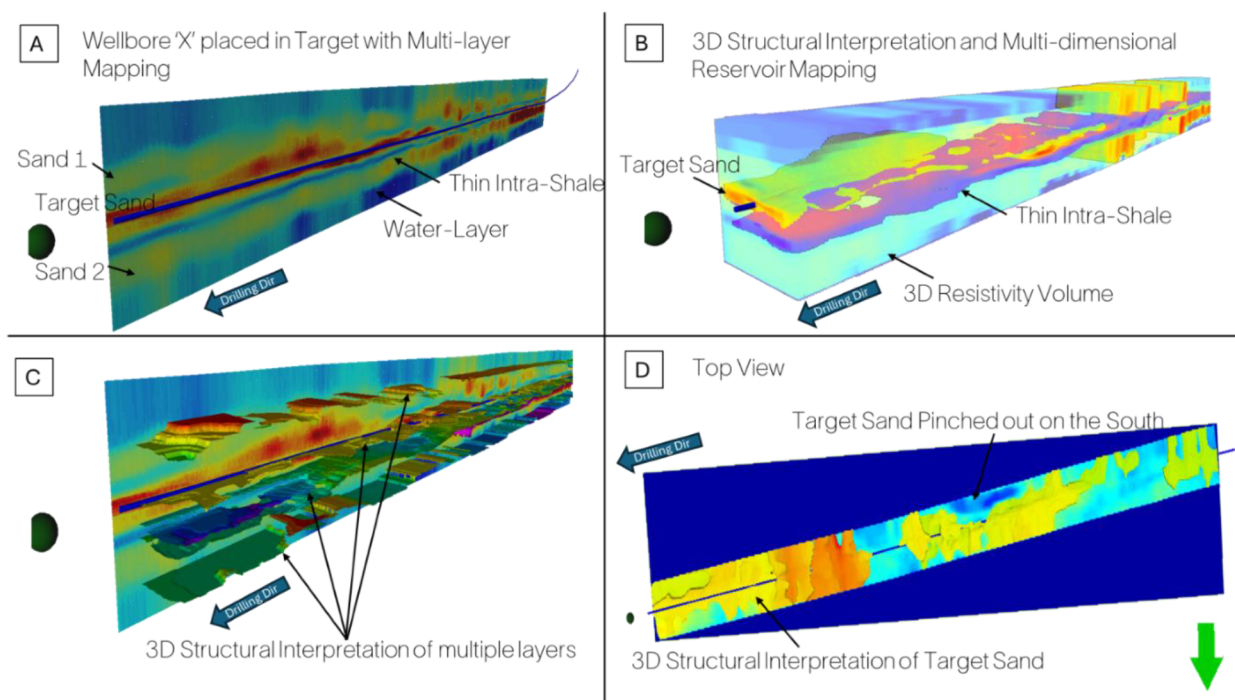


Figure 4—A) Well 'X' placed in the target sand and UDAR tool mapping multiple layers of the reservoir; B) 3D Reservoir Mapping Answer products highlighting the thin intra-shale below the wellbore and the pay zone. C) 3D Structural Interpretation of the difference layers of the reservoir providing true Reservoir Characterization. D) Well 'X' was few feet north of the target sand pinching out.

Multi-Dimensional Mapping for Intelligent Well Placement. Building on the successful placement of wellbore X and the insights gained from multidimensional reservoir mapping, a follow up lateral wellbore Y, was planned to target a deeper sand layer identified through the resistivity mapping inversions of wellbore X. A critical input to the reservoir characterization workflow was the 3D structural interpretation of multiple reservoir layers, which were used to update the existing grids to gain a better understanding of the dynamic and changing nature of these fluvial sand channels. One notable observation was that the target sand in wellbore X exhibited a pinch out toward the southern side of the wellbore. Consequently, the wellbore Y was deliberately designed to avoid the same sand and instead target a deeper sand layer below. In Figure 5C, the projection of the updated grid interpretations from wellbore X can be seen on the intersection plane of wellbore Y. The planned trajectory, in purple, is seen to be placed in the sand '2' just above the potential water, with a certain cutoff used as the no-go limit.

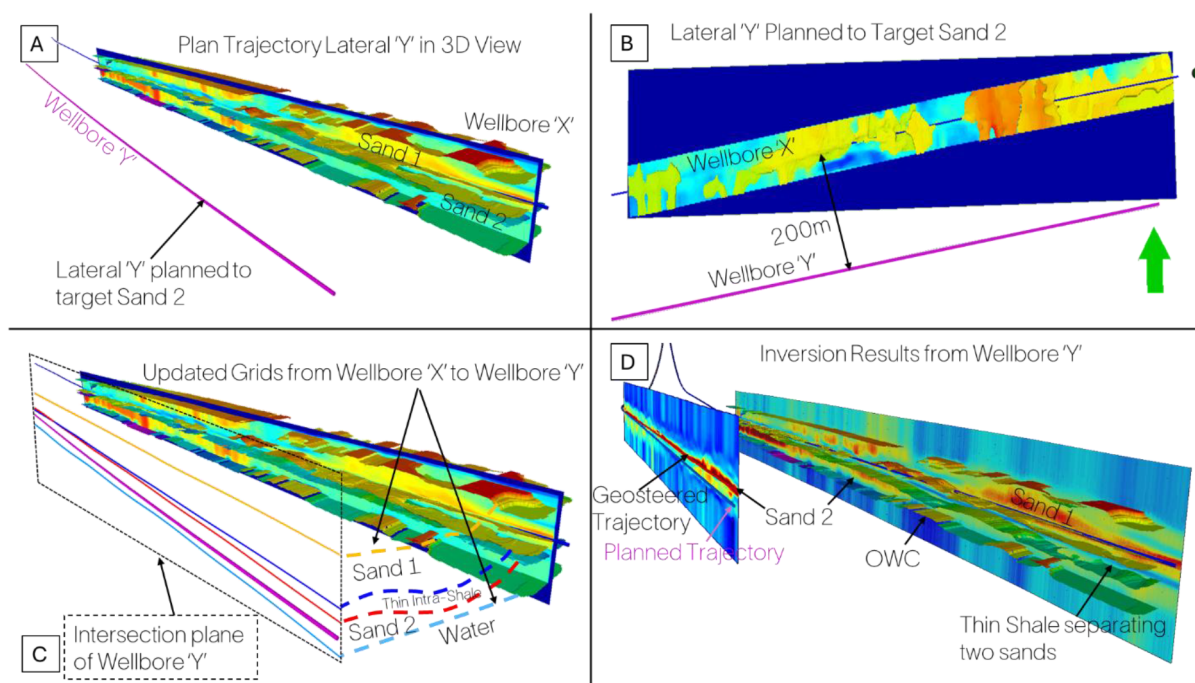


Figure 5—A) Wellbore Y planned in the deeper than Wellbore X to target Sand 2; B) Map View showing distance between the planned lateral ‘Y’ and already drilled Wellbore ‘X’; C) 3D Structural Interpretation of the different layers of the reservoir projected on the Wellbore ‘Y’ intersection plane; D) Inversion results from Wellbore ‘Y’ confirming the expected Sand depth and Water Contact.

The inversion results from wellbore Y validated the expected target sand positioning and confirmed prior structural interpretations from wellbore X. Wellbore Y, was successfully landed and maintained with sand 2 along the drilled interval. However, real time geosteering was essential to maintain a certain stand-off distance from the mapped oil water contact. In Figure 5D, the planned trajectory is shown in purple and the geosteering trajectory in black. The geosteered trajectory is shallower than the planned one, showing the necessary standoff maintained from the oil water contact mapped below the wellbore and also not exiting the target sand into the shale layer above. This example clearly demonstrates the value of integrating UDAR reservoir mapping data from a single wellbore to inform placement of nearby wells. In this case, the distance between wellbores X and Y, as seen in Figure 5B, was approximately 200 meters. The correlation between nearby wells is quite dependent on the distance between them, although there is no empirically established relationship the general understanding is that the farther apart the wellbores are, the lower the chances of achieving a good match between results—particularly in a complex depositional system such as the one presented in this paper

Proactive Completion Optimization. The completion design of another well (Well-M) represents an advanced application of UDAR in smart reservoir management. The integration of UDAR 1D resistivity inversions and 2D transverse resistivity inversions from UDAR, in combination with shallower LWD measurements provided valuable insights into the reservoir heterogeneity and variation in water contact. The primary objective was to reduce uncertainty in the structural framework and saturation profile, thereby enhancing our understanding of the reservoir's dynamics and informing optimal completion strategies. The UDAR resistivity inversion data, which provided mapping of formation boundaries based on resistivity contrasts up to 80 ft TVD from the wellbore. Both longitudinal and transversal structural interpretations assisted in adjusting the well trajectory to place it in the reservoir's sweet spot. The 2D Transverse resistivity mapping and 3D resistivity volumes from UDAR identified future potential water breakthrough zones into the wellbore, forming a crucial part of the production strategy for this well.

In the UDAR inversion process, the well M initially encounters a low resistivity zone (blue color), interpreted as low-quality sand, designated as Zone A in the heel section of the well. It then transitions into the resistive reservoir facies (red). As the well continues drilling through the resistive reservoir layers towards the toe, it targets the thick merged section of multiple resistive facies. Midway through the drilled section (Zone B), UDAR resistivity inversions begin mapping a conductive boundary slightly higher and closer to the wellbore (Figure 6). As drilling progresses, the conductive zone moves even closer in Zone C until the wellbore enters the thick resistive target formation. Upon entry, LWD logs was unable to detect the proximity to the water along the wellbore to make informed decision about the intelligent completion. The presence of water was only understood after integrating UDAR answer products with LWD data interpretation. To prolong well life and recovery from this well by delaying the water production, a well-planned completion design was implemented to avoid producing water early from these zones.

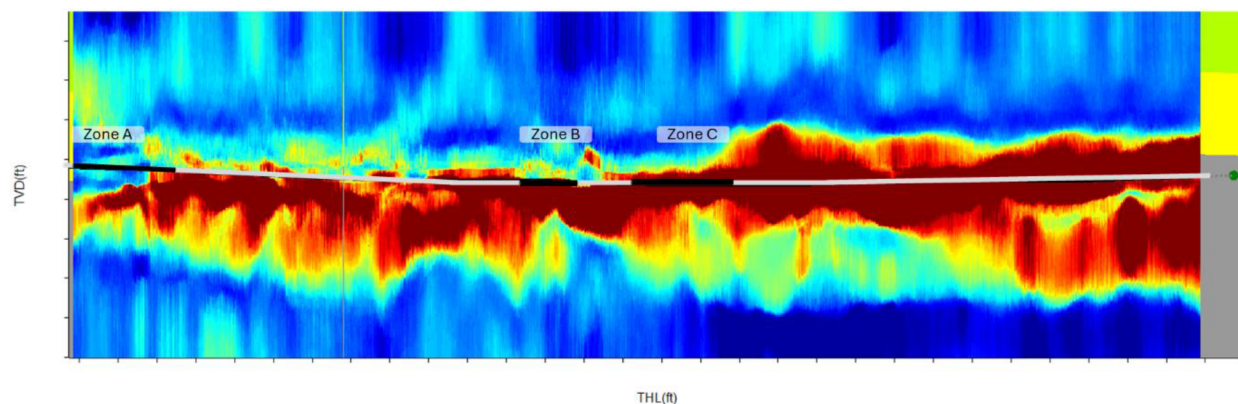


Figure 6—Completion optimization in well M where zones A, B and C were avoided in production to avoid the shallow aquifer.

Subsurface History Matching Validation. Reservoir history matching is a calibration process in which reservoir simulation models are iteratively adjusted to reproduce observed pressure and production data. This involves modifying geological and dynamic parameters to reduce mismatch between simulated and measured responses, thereby enhancing the model's predictive reliability. However, history matching can be an inherently non-unique inverse problem, where multiple reservoir model realizations can match the observed production trends. This uncertainty often arises due to limited subsurface constraints, resulting in models that may match rates and pressures but fail to capture true underlying reservoir dynamics. To improve model reliability, it is essential to calibrate both surface responses and subsurface fluid condition.

UDAR data offers deep-reading resistivity measurements that enhance the mapping of fluid contacts and saturation profiles beyond the immediate wellbore vicinity. When integrated into the history matching workflow, UDAR data provides additional constraints on the vertical and areal distribution of fluids, particularly in regions with limited surveillance.

In the example shown in Figure 7A, the dynamic model was initialized to match the Free Water Level (FWL). As production advanced, factors such as aquifer influx, permeability heterogeneity, and mobility contrasts contributed to the displacement of the water contact. Conventional history matching, relying on adjustments to relative permeability, transmissibility multipliers, and saturation endpoints yielded acceptable matches for field production rates and bottomhole pressures. However, the resulting water saturation distribution remained highly uncertain.

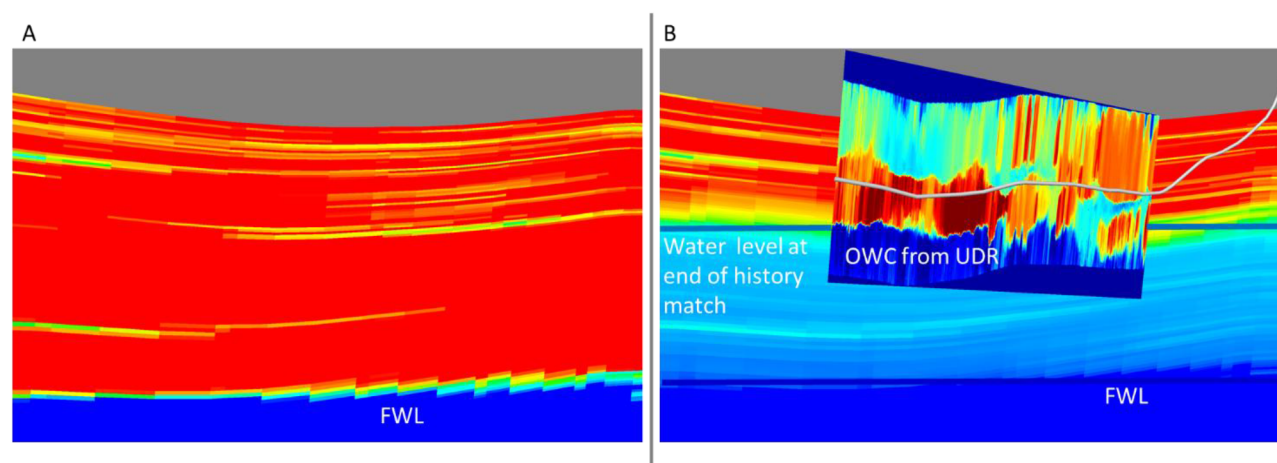


Figure 7—A) Model water saturation at initial time step with water level at free water level (FWL) B) Model water saturation at end of history match with shallow water level and matching the OWC interpreted from UDAR in recent well

Subsequent acquisition of UDAR data in a nearby development well provided direct measurement of the water contact at the end of the history period. The simulated water saturation at that time step, shown in Figure 7B, closely matched with the UDAR-derived contact, validating the model's dynamic consistency.

By incorporating UDAR inverted data, the model calibration is extended beyond surface data to include subsurface fluid distribution post production and end of history. This reduces uncertainty, enhances predictive capability, and establishes a more robust foundation for infill well planning, reserves estimation, and overall reservoir management.

4D Time-Lapse Surveillance Model Enhancement. An advanced reservoir surveillance technique integrates petrophysical logs with time-lapse data to construct a 4D model that captures fluid movement both spatially and over time. This workflow utilizes a geostatistical distribution algorithm to simulate water saturation changes at discrete intervals, while honoring reservoir structure and heterogeneity.

Conventional inputs such as open-hole logs, Pulsed Neutron Logs (PNL), and Production Logging Tool (PLT) data provide insights into saturation trends but are often limited by sparse spatial coverage and acquisition frequency. These limitations constrain model resolution and reliability.

The incorporation of UDAR inversion data enhances model reliability by introducing additional areal constraints on oil-water contact along the wellbore trajectory. Pseudo-wells constructed from UDAR-derived water saturation profiles increase spatial data density, improving the accuracy of tracking dynamic water contact movements.

Example shown in Figure 8 highlights the improvement in the baseline model's final water saturation map with the help of offset well UDAR data. PNL data from well C suggests water encroachment from the east, but well D drilled just before the final time step reveals a higher water level than predicted. By integrating Well D UDAR-derived saturation profile into the model, the updated result (Figure 8B) captures the actual water movement more accurately.

Integrating UDAR inverted data into 4D modeling workflows improves calibration, supports dynamic history matching, and reduces uncertainty in characterizing water encroachment trends. This improves reservoir forecasting accuracy and supports more effective field development planning.

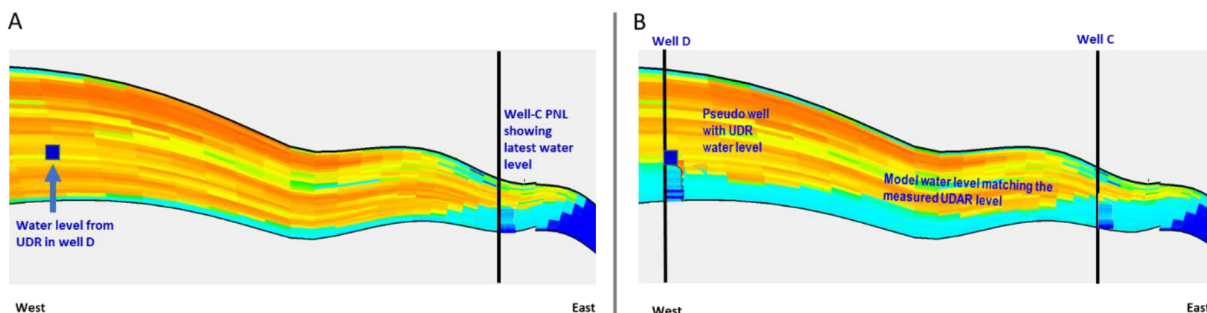


Figure 8—A) Model water saturation showing water encroachment from East but only reaching up to well C. Well D showing water reached much higher-level B) Model water saturation after incorporating UDAR data point in the modeling workflow, water level now matches the actual water level seen in the well D

Conclusion

In summary, this study demonstrates that UDAR technology is a game-changer not only in precise geosteering but as a powerful tool for post-drilling reservoir management. The integration of high-resolution, multidimensional resistivity data has proven essential for unlocking deeper insights into reservoir characterization, fluid contacts, and dynamic behavior. These capabilities enable smarter well placement and more reliable model calibration, ultimately enhancing surveillance, and significantly mitigating uncertainty and development risk.

Looking ahead, the future of UDAR technology is poised for even greater impact. Advances in inversion methodologies, real-time data analytics, and the fusion of UDAR with machine learning and digital twin frameworks promise to elevate reservoir modeling from static snapshots to continuously updated, predictive tools. This evolution will empower operators to make faster, more informed decisions, optimize recovery strategies, and extend field life in complex and challenging reservoirs.

By fully leveraging UDAR's potential, the industry can drive a paradigm shift towards smarter, more sustainable resource management unlocking value that was previously inaccessible and setting a new standard for reservoir surveillance and development excellence.

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